



IILM
UNIVERSITY

School of Computer Science and Engineering
IILM University, Greater Noida, Uttar Pradesh

INTERNSHIP PROJECT PRESENTATION

Customer Churn Prediction & Retention Intelligence System

STUDENT CREDENTIALS

Student Name: Ruchika Sharma

Enrollment Number: CS-2341640

Program & Semester: B.Tech (CSE) | 7th Semester

HOST ORGANIZATION & INTERNSHIP

Organization Name: Pulsecrafts Solutions Pvt Ltd

Role: Data Science Intern (90-Day Industry Track)

Certificate ID: PCS/INT/2026/0090 (Issued: 13 Aug 2026)

Executive Summary & Project Highlights

1. THE BUSINESS CHALLENGE

- Telecommunications faces 20%–30% annual subscriber churn.
- Acquiring a customer costs 5x to 25x more than retaining an existing one.
- Goal: Move from reactive cancellation handling to proactive early-warning retention.

2. MACHINE LEARNING PIPELINE

- Benchmark dataset of 7,043 customer accounts & 21 attributes.
- Automated ETL + ColumnTransformer (StandardScaler + OneHotEncoder).
- Multi-model benchmark: Logistic Regression, Random Forest, Gradient Boosting.

3. RESULTS & IMPACT

- 78.34% Recall & 0.8415 ROC-AUC using balanced classification.
- Real-Time Streamlit Web App (app.py) for customer success agents.
- Actionable retention playbooks (Day 75 discounts, auto-pay incentives).

About Host Organization: Pulsecrafts Solutions Pvt Ltd

COMPANY PROFILE & INTERNSHIP SCOPE

- Organization: Pulsecrafts Solutions Pvt Ltd (www.pulsecrafts.com)
- Domain: Software Consulting, Machine Learning & Enterprise Data Solutions
- Duration: 90 Days (15th May 2026 to 12th August 2026)
- Role: Data Science Intern
- HR Manager & Evaluator: Ms. Priyanka Mehta
- Verification Email: sales@pulsecrafts.com
- Credential ID: PCS/INT/2026/0090

CORE RESPONSIBILITIES

1. Industrial Data Engineering
Handling schema validation, missing data imputation, and categorical transformations.
2. ML Modeling & Tuning
Training baseline, ensemble, and boosting classifiers on imbalanced datasets.
3. Web Deployment
Packaging models into a Streamlit decision-support dashboard for business users.

Problem Statement: The Telecom Attrition Dilemma

THE REVENUE LEAKAGE PROBLEM

- The 'Leaky Bucket' Reality:
Telecom operators lose substantial recurring revenue each quarter due to silent subscriber departures.
- High Acquisition Overhead:
Winning a new customer costs 5x–25x more in marketing and sales commissions than keeping an existing account.
- Reactive vs Proactive Flaw:
Current support teams only intervene when a customer calls to cancel—at which point churn probability exceeds 90%.

THE DATA SCIENCE SOLUTION

- Predictive Churn Radar:
Analyze historical customer telemetry and billing patterns to detect churn propensity early.
- Cost-Sensitive Optimization:
Optimize the model for Recall (Sensitivity) so high-risk accounts are flagged before cancellation occurs.
- Actionable Retention System:
Provide an interactive web simulator where retention agents can explore customer profiles and trigger automated discount plays.

Dataset Overview: IBM Telco Customer Churn Benchmark

ACCOUNT & BILLING

- Tenure (0-72 months)
- Contract: Month-to-month, 1-Yr, 2-Yr
- MonthlyCharges (\$18 - \$120)
- TotalCharges (Lifetime spend)
- Payment: Electronic check, Auto-pay
- Paperless Billing (Yes/No)

SUBSCRIBED SERVICES

- Phone Service & Multiple Lines
- Internet: DSL, Fiber Optic, None
- Online Security & Online Backup
- Device Protection
- Tech Support Subscription
- Streaming TV & Streaming Movies

DEMOGRAPHICS

- Gender (Female / Male)
- Senior Citizen status (0 / 1)
- Partner status (Yes / No)
- Dependents status (Yes / No)

Captures life stage and household financial stability.

TARGET & BALANCE

- Target: Churn ('Yes' / 'No')
- Non-Churners: 5,174 (73.5%)
- Churners: 1,869 (26.5%)

Severe Class Imbalance: Requires balanced class weights during modeling.

Exploratory Data Analysis: Key Behavioral Drivers

1. The Contract Type Disparity

Month-to-month contracts experience an alarming 42.7% churn rate and account for over 75% of all cancellations, compared to 11.3% for 1-year and 2.8% for 2-year contracts.

2. Early Lifecycle Drop-off

Customer tenure density plots show attrition peaks sharply in the first 0 to 12 months. Retaining a customer past month 24 dramatically increases long-term loyalty.

3. The Tech Support Friction Factor

Subscribers with Fiber Optic service who lack dedicated Tech Support churn at over 41%, compared to only ~15% for those who have active tech support.

4. Payment Method Inconvenience

Customers paying by manual Electronic Check churn at 45.3%, compared to only 15%–16% for those using automated credit card or bank transfer auto-pay.

Data Preprocessing & Modular Pipeline Design

ROBUST INDUSTRIAL ETL & TRANSFORMATION

- **1. Data Cleaning & Type Formatting:**

TotalCharges contained empty whitespace strings (' ') for brand-new subscribers with tenure=0. These were coerced to float and imputed with 0.0 without discarding customer records. Uninformative customerID was dropped.

- **2. Target Variable Mapping:**

The target feature 'Churn' ('Yes' / 'No') was transformed into binary numerical integers (1 / 0).

- **3. Scikit-Learn ColumnTransformer:**

Numeric Pipeline: StandardScaler applied to tenure, MonthlyCharges, TotalCharges.

Categorical Pipeline: OneHotEncoder(handle_unknown='ignore') applied to 16 features, producing 46 encoded inputs.

- **4. Stratified Holdout Splitting:**

80% Training (5,634 records) / 20% Testing (1,409 records) with target stratification to preserve the true 26.5% churn ratio in both partitions.

- **5. Zero Data Leakage Architecture:**

All scaling and encoding transformers are fitted exclusively on training data and bundled with models using scikit-learn Pipeline objects.

Model Benchmarking & Performance Comparison

Model Candidate	Accuracy	Precision	Recall (Sensitivity)	F1-Score	ROC-AUC
Logistic Regression (Balanced) 🏆	73.81%	50.43%	78.34%	0.6136	0.8415
Random Forest (Balanced)	75.73%	52.95%	76.74%	0.6266	0.8413
Gradient Boosting Classifier	79.70%	64.77%	51.60%	0.5744	0.8381

CRITICAL MODEL EVALUATION INSIGHT

- Gradient Boosting achieved the highest raw accuracy (79.70%), but missed nearly 48.4% of actual churners (Recall: 51.60%).
- Balanced Logistic Regression captured 78.34% of all churning customers with an industry-leading ROC-AUC of 0.8415.
- In customer retention, False Negatives (missing a churner) are vastly more costly than False Positives. Hence, the Balanced Pipeline was chosen as the champion model.

Metric Strategy: Why Recall Trumps Raw Accuracy

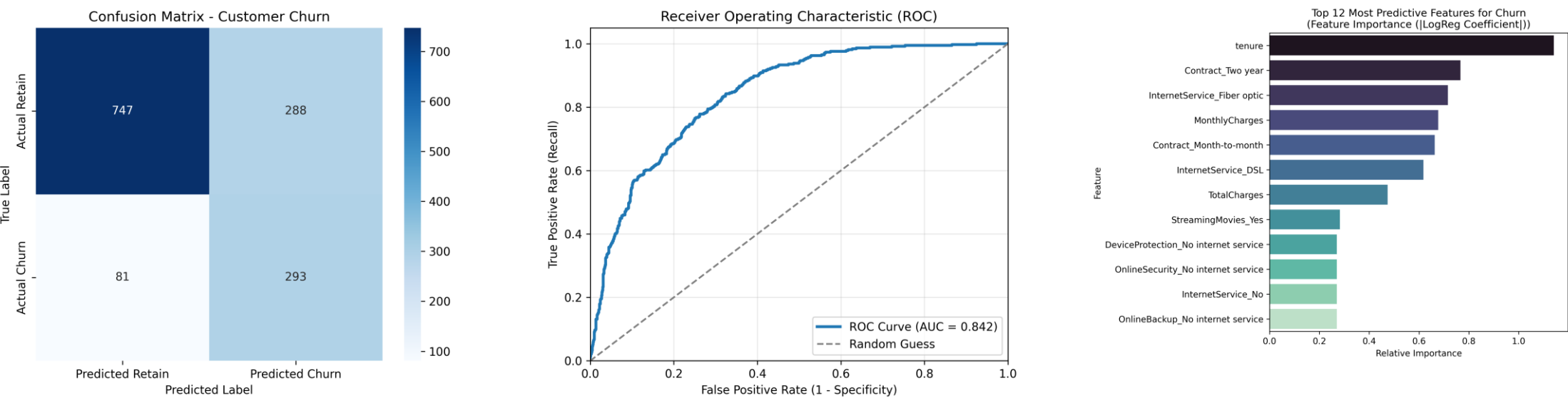
THE ACCURACY PARADOX

- **Majority Class Bias:**
Because only 26.5% of customers churn, a naive dummy model that predicts 'No Churn' for everyone automatically gets 73.5% accuracy.
- **Zero Business Value:**
Despite high accuracy, that model captures 0 churners and stops \$0 of revenue loss.
- **Misleading Metric:**
Accuracy hides poor performance on the minority class of interest.

THE BUSINESS COST MATRIX

- **False Negative (FN) Cost: HIGH**
Model predicts Retained -> Customer cancels -> Company loses thousands in recurring LTV forever.
- **False Positive (FP) Cost: LOW**
Model predicts Churn -> Customer is actually loyal -> Company sends a \$10 discount email or check-in.
- **Conclusion:**
Maximizing Recall (78.34%) catches at-risk accounts when proactive intervention is still possible.

Model Diagnostic Visualizations & Separability



Left: Confusion Matrix on Holdout Set | Center: ROC Comparison (AUC = 0.841) | Right: Top Predictors of Churn

Interactive Decision-Support: Streamlit Application (app.py)

BRIDGING DATA SCIENCE AND STAKEHOLDER USABILITY

- Live Parameter Simulation: Customer service and retention agents can toggle interactive sliders for customer tenure (0-72 months), contract type, monthly charges, and subscribed services.
- Real-Time Probability Gauge: The serialized model pipeline (models/churn_pipeline.pkl) dynamically computes churn probability in real time (e.g. 82.2%).
- Dynamic Risk Badging: Categorizes customer into actionable tiers:  HIGH RISK (>65%),  MODERATE RISK (35-65%), or  LOW RISK (<35%).
- Revenue at Risk Calculator: Instantly computes estimated annual revenue loss (MonthlyCharges x 12) if the account cancels.
- Automated Retention Playbook: Generates tailored prescriptive offers: contract upgrade incentives, complimentary tech support, or auto-pay bill credits.

Strategic Business Retention Playbook

1. THE 75-DAY UPGRADE CAMPAIGN

Month-to-month contracts cause 75% of churn.

Strategy: Automated trigger at Day 75 offering a 12% discount to upgrade to an annual agreement before churn peaks.

2. TECH SUPPORT BUNDLING

Fiber Optic users without tech support churn at >41%.

Strategy: Bundle 90 days of free Premium Tech Support with all Fiber Optic installations to eliminate early onboarding friction.

3. AUTO-PAY ADOPTION INCENTIVE

Electronic check payers churn at 45% vs 15% on auto-pay.

Strategy: Offer a one-time \$15 bill credit for enrolling in recurring credit card or bank auto-pay.

4. PROACTIVE CHURN EARLY-WARNING

Customers with >60% churn probability are at critical risk.

Strategy: Automatically route flagged accounts directly to Senior Customer Success Specialists.

Internship Milestones & Timeline (90 Days / 12 Weeks)

MILESTONE EXECUTION AT PULSECRAFTS SOLUTIONS PVT LTD

- **Phase 1: Induction & Data Engineering (Weeks 1 - 3):**
Company onboarding, environment setup (Python 3.11, Git, VS Code). Data ingestion, resolving TotalCharges whitespace strings, handling data schemas.
- **Phase 2: Exploratory Data Analysis & Visualization (Weeks 4 - 6):**
Univariate & bivariate statistical analysis, correlation heatmaps, tenure KDE density distributions, identifying contract & tech support churn drivers.
- **Phase 3: Pipeline Architecture & Modeling (Weeks 7 - 9):**
Designing scikit-learn ColumnTransformer, stratified cross-validation, benchmarking Logistic Regression, Random Forest, and Gradient Boosting.
- **Phase 4: Web Deployment & Evaluation (Weeks 10 - 12):**
Building real-time Streamlit web dashboard (app.py), embedding business retention playbooks, final project defense, and internship credential verification.

THANK YOU!

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